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Investor Attention and Lead-Lag Networks in U.S. Equities

1. Hypothesis Pitch

Many companies listed on the stock market that operate within the same industry are often exposed to common economic developments, including changes in demand, financing conditions and regulation. However, information may not always be reflected in these companies' stock prices simultaneously. Stocks receiving greater investor attention may attract more immediate analysis and trading activity, while investors may take longer to recognise how the same information affects less-followed companies (Brennan, Jegadeesh and Swaminathan, 1993; Hou, 2007). The difference in how quickly information is processed could create a short-lived opportunity to anticipate price adjustments in less-followed stocks.

This project investigates whether investor attention can improve stock return predictions using lead-lag relationships between same-industry U.S. stocks. Specifically, when a high-attention stock experiences an unusually large daily return, a lower-attention stock in the same industry is expected to respond on the next trading day in the direction suggested by their historical lead-lag relationship. Thus, focusing on relationships where the leader has higher attention than the follower should produce better trading results than selecting relationships based only on past price movements.

To test this hypothesis, an attention-informed lead-lag network will be compared to a simple price-based lead-lag network. If the attention-informed network does not consistently outperform the price-only network across reasonable parameter choices, there would be insufficient evidence that news attention adds predictive value. Furthermore, if any apparent advantage disappears after accounting for transaction costs, the proposed trading opportunity would lack economic viability (Novy-Marx and Velikov, 2016).

2. Data Pipeline

2.1 Data source and acquisition

The data was retrieved through Bloomberg's authenticated Excel Add-in, using Spreadsheet Builder to generate BDH formulas for the selected securities, data fields and date range. The obtained observations were then saved as fixed-value Excel snapshots for use in Python. The requested period spanned from 1 January 2020 to 30 September 2026, providing several years of daily observations across different market conditions, including the COVID-19 disruption and subsequent changes in inflation, interest rates and market volatility. It also provides sufficient historical observations for rolling estimation and chronological evaluation. The dataset contained 50 large U.S.-listed equities across five industries: Banks, Technology, Consumer, Healthcare and Energy, giving 450 possible directed same-industry relationships.

The individual equities were selected manually from established U.S.-listed companies rather than through a formal market-capitalisation or liquidity ranking. Their suitability was then assessed using 2020-2023 news coverage and average daily dollar trading volume, to ensure that the sample contained meaningful differences in attention and sufficient trading activity. Average daily news counts over 2020-2023 ranged from 36 (MPC) to 2,099 (AAPL), and the lowest average daily dollar volume was about $187 million (STT).

Daily opening prices were later obtained through a separate Bloomberg Excel request, giving a total of five Bloomberg data fields, each serving a distinct purpose in investigating the hypothesis:

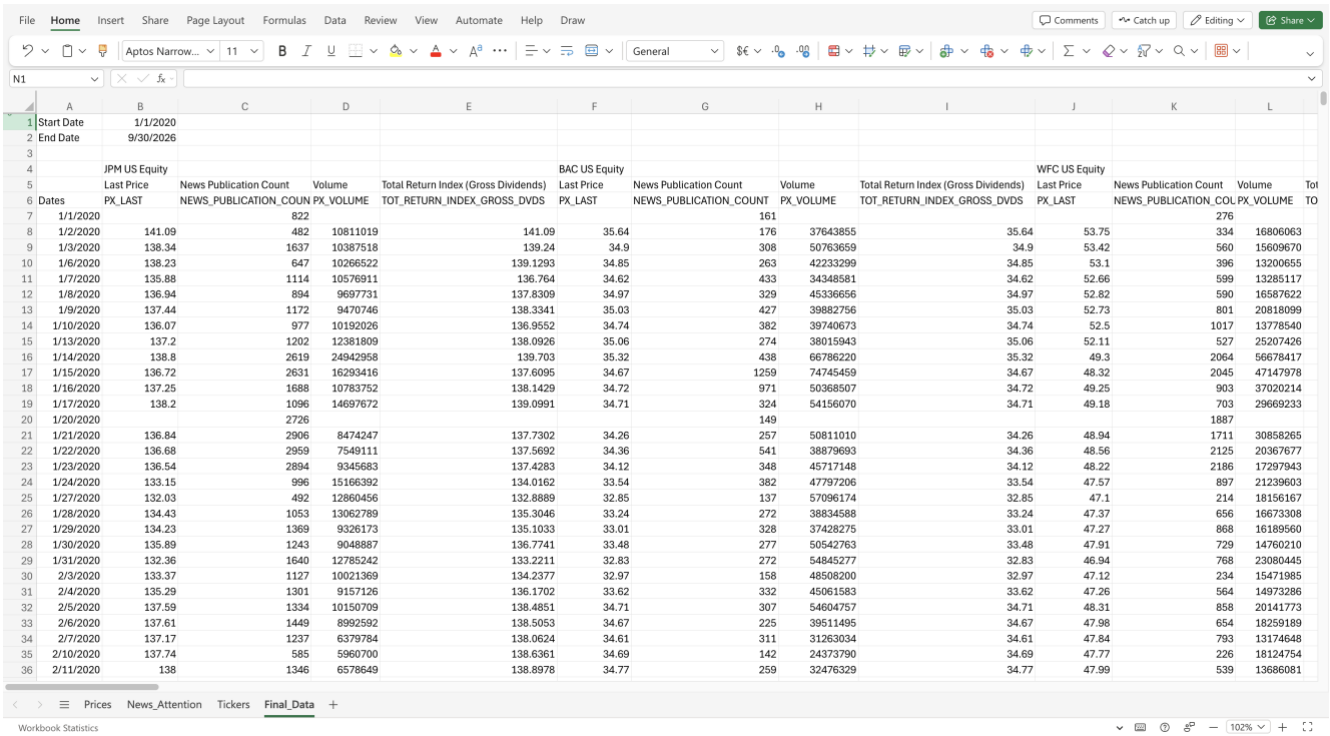
Bloomberg field	Purpose and justification
PX_LAST	Daily closing prices used to calculate stock returns and identify lead-lag relationships.
NEWS_PUBLICATION_COUNT	Daily financial-news publication counts, providing an observable proxy for differences in investor attention.
PX_VOLUME	Daily trading volume used to assess historical trading activity within the selected stock universe, but not included in the trading signal.
TOT_RETURN_INDEX_GROSS_DVDS	Gross total-return index used to evaluate strategy performance, including dividends rather than relying only on changes in share prices.
PX_OPEN	Daily opening prices used to test strategy performance when trades are entered at the following day's open rather than the signal day's close.

Table 2.1: Bloomberg fields and their purpose

Company fundamentals and macroeconomic indicators were outside the scope of this study, as it primarily focuses on short-horizon relationships between stock returns and news coverage.

2.2 Data cleaning and transformation

Figure 2.1 shows the raw Bloomberg spreadsheet containing multiple data fields for each security, arranged in repeated column groups rather than as a conventional analysis-ready table. Python's pandas library was used to extract observation dates, identify securities, convert data fields to numeric values and organise the observations into separate price, news, volume and total-return tables. Date formats were standardised, including the conversion of Excel serial dates, and chronological ordering was checked before calculating returns.



	A	B	C	D	E	F	G	H	I	J	K	L
1	Start Date	1/1/2020										
2	End Date	9/30/2026										
3												
4		JPM US Equity				BAC US Equity				WFC US Equity		
5		Last Price	News Publication Count	Volume	Total Return Index (Gross Dividends)	Last Price	News Publication Count	Volume	Total Return Index (Gross Dividends)	Last Price	News Publication Count	Volume
6	Dates	PX_LAST	NEWS_PUBLICATION_COUN	PX_VOLUME	TOT_RETURN_INDEX_GROSS_DVDS	PX_LAST	NEWS_PUBLICATION_COUNT	PX_VOLUME	TOT_RETURN_INDEX_GROSS_DVDS	PX_LAST	NEWS_PUBLICATION_COU	PX_VOLUME
7	1/1/2020		822				161				276	
8	1/2/2020	141.09	482	10811019	141.09	35.64	176	37643855	35.64	53.75	334	16806063
9	1/3/2020	138.34	1637	10387518	139.24	34.9	308	50763659	34.9	53.42	560	15609670
10	1/6/2020	138.23	647	10266522	139.1293	34.85	263	42233299	34.85	53.1	396	13200655
11	1/7/2020	135.88	1114	10576911	136.764	34.62	433	34348581	34.62	52.66	599	13285117
12	1/8/2020	136.94	894	9697731	137.8309	34.97	329	45336656	34.97	52.82	590	16587622
13	1/9/2020	137.44	1172	9470746	138.3341	35.03	427	39882756	35.03	52.73	801	20818099
14	1/10/2020	136.07	977	10192026	136.9552	34.74	382	39740673	34.74	52.5	1017	13778540
15	1/13/2020	137.2	1202	12381809	138.0926	35.06	274	38015943	35.06	52.11	527	25207426
16	1/14/2020	138.8	2619	24942958	139.703	35.32	438	66786220	35.32	49.3	2064	56678417
17	1/15/2020	136.72	2631	16293416	137.6095	34.67	1259	74745459	34.67	48.32	2045	47147978
18	1/16/2020	137.25	1688	10783752	138.1429	34.72	971	50368507	34.72	49.25	903	37020214
19	1/17/2020	138.2	1096	14697672	139.0991	34.71	324	54156070	34.71	49.18	703	29669233
20	1/20/2020		2726				149				1887	
21	1/21/2020	136.84	2906	8474247	137.7302	34.26	257	50811010	34.26	48.94	1711	30858265
22	1/22/2020	136.68	2959	7549111	137.5692	34.36	541	38879693	34.36	48.56	2125	20367677
23	1/23/2020	136.54	2894	9345683	137.4283	34.12	348	45717148	34.12	48.22	2186	17297943
24	1/24/2020	133.15	996	15166392	134.0162	33.54	382	47797206	33.54	47.57	897	21239603
25	1/27/2020	132.03	492	12860456	132.8889	32.85	137	57096174	32.85	47.1	214	18156167
26	1/28/2020	134.43	1053	13062789	135.3046	33.24	272	38834588	33.24	47.37	656	16673308
27	1/29/2020	134.23	1369	9326173	135.1033	33.01	328	37428275	33.01	47.27	868	16189560
28	1/30/2020	135.89	1243	9048887	136.7741	33.48	277	50542763	33.48	47.91	729	14760210
29	1/31/2020	132.36	1640	12785242	133.2211	32.83	272	54845277	32.83	46.94	768	23080445
30	2/3/2020	133.37	1127	10021369	134.2377	32.97	158	48508200	32.97	47.12	234	15471985
31	2/4/2020	135.29	1301	9157126	136.1702	33.62	332	45061583	33.62	47.26	564	14973286
32	2/5/2020	137.59	1334	10150709	138.4851	34.71	307	54604757	34.71	48.31	858	20141773
33	2/6/2020	137.61	1449	8992592	138.5053	34.67	225	39511495	34.67	47.98	654	18259189
34	2/7/2020	137.17	1237	6379784	138.0624	34.61	311	31263034	34.61	47.84	793	13174648
35	2/10/2020	137.74	585	5960700	138.6361	34.69	142	24373790	34.69	47.77	226	18124754
36	2/11/2020	138	1346	6578649	138.8978	34.77	259	32476329	34.77	47.99	539	13686081

Figure 2.1: Structure of the Bloomberg export (Final_Data sheet), with four fields per security in repeated column groups

Missing observations were examined by security, data field and date. Of the 1,761 dates in the Bloomberg export, 66 were excluded because no company had a valid closing price; these correspond to U.S. market holidays, leaving 1,695 trading dates across all 50 equities. Two missing news-publication observations were retained as missing rather than replaced with zero, since unavailable data does not necessarily indicate that no news was published.

Investor attention was approximated using a 20-trading-day rolling average of Bloomberg's daily news-publication counts. The measure was shifted by one trading day to avoid using current-day news when selecting relationships.

Corporate actions were checked by comparing daily price returns with total-return-index returns, which are unaffected by splits. The largest daily gap was 2.9%, on Occidental Petroleum's 53% fall on 9 March 2020, consistent with a dividend rather than a split, and no other one-day price move exceeded 40%, so the closing prices contain no unadjusted splits.

2.3 Data Limitations and Reproducibility

Since the universe contains only large, established companies, the findings may not apply to smaller or less-followed firms, and the fixed list creates survivorship bias (see Section 5.2). The industry groups are also relatively broad and may contain companies with different business activities, making it harder to determine whether historical relationships reflect shared economic information. Moreover, the first development networks were estimated almost entirely on 2020 data, where COVID-19 volatility produced strong and mostly negative lagged correlations, so early development signals reflect crisis-period relationships.

Both Bloomberg data snapshots, containing the original price, news, volume and total-return fields and the separately downloaded opening prices, are included with the Jupyter notebook. The notebook automatically reads these files, checks dates and missing observations, constructs the model inputs and reproduces the analysis without requiring Bloomberg Terminal access.

3. Strategy and Backtest

3.1 Strategy

Historical lead-lag relationships were identified by examining whether a stock's daily return was associated with another stock's return on the following trading day. Relationships are calculated from the stock returns of the previous 252 trading days (around 1 calendar year), and form a network that is rebuilt every 20 trading days. Within each industry, relationships are ranked by the absolute correlation between a stock's return and the other stock's return on the next day. The 18 strongest relationships (20%) from each industry are used to generate trading signals.

A trading signal is generated when a leading stock experiences an unusually large daily return, defined as a return z-score of at least 2 in absolute value relative to its previous 20 trading days. A position is then taken in the follower at that day's close and held for one trading day, with its direction determined by whether the historical relationship indicated continuation or reversal.

The two strategies evaluated in this study only differ in how relationships are selected. The price-only network, which acts as the baseline, selects relationships using historical returns alone. The attention-informed network requires the leading stock to have higher 20-day average Bloomberg news coverage than the follower at each rebalance. Both networks kept 18 relationships per industry, so any difference comes from which relationships were selected, not how many.

3.2 Backtesting

Trading performance was first assessed over the 2021-2023 development period, followed by the January 2024-September 2026 out-of-sample period. Performance was calculated using Bloomberg's

gross total-return data and measured through cumulative return, Sharpe ratio (annualised from daily returns with a zero risk-free rate) and maximum drawdown. Results were evaluated before and after transaction costs, assuming a cost of 10 basis points (0.10%) on each purchase or sale.

Specification	Cumulative return	Sharpe ratio	Maximum drawdown
Development: price-only (gross)	−7.36%	−0.165	−21.86%
Development: attention (gross)	−8.08%	−0.194	−26.00%
Development: price-only (net, 10 bps)	−52.65%	-2.113	-54.37%
Development: attention (net, 10 bps)	-49.60%	-1.971	-52.77%
Later: price-only (gross)	+29.40%	0.848	−9.94%
Later: attention (gross)	+24.03%	0.815	−10.30%
Later: price-only (net, 10 bps)	−33.13%	−1.170	−33.88%
Later: attention (net, 10 bps)	−30.68%	−1.248	−31.08%

Table 3.1: Key results, close-to-close execution

Both strategies lost money during the development period, but achieved positive returns in the later period. Before costs, the price-only strategy outperformed the attention-informed strategy in both periods, providing insufficient evidence that news attention consistently improves trading performance. To test whether this gap was meaningful, a moving-block bootstrap of the later-period daily returns was used, resampling 20-day blocks (5,000 resamples) so that the dependence between consecutive days is preserved. The 95% confidence interval for the mean daily difference between the attention-informed and price-only strategies was −0.040% to +0.026%, which includes zero, so the two strategies cannot be statistically distinguished. The intervals for each strategy's own mean gross daily return also included zero, so even the gross later-period profits were not significant at the 5% level.

The change between the development and later periods suggest that the predictive value of these historical relationships was not consistent over time, although the results do not establish what caused this variation. The gross Sharpe ratios of 0.848 and 0.815 indicate positive but moderate risk-adjusted performance, contrasting the negative Sharpe ratios during development. In the later period the worst gross peak-to-trough loss was about 10%, close to one year's volatility (12.0% and 10.3% annualised), whereas in development both strategies at some point lost more than a fifth of their peak value. After costs, both strategies lost money in both periods, and net maximum drawdowns almost equal cumulative losses, showing that costs depleted capital steadily from the start rather than through isolated losing intervals.

An additional test, run on the later evaluation period only, used opening prices to examine performance when trades were entered at the following day's open rather than at the signal-day close.

Specification	Cumulative return	Sharpe ratio	Maximum drawdown
Price-only (gross)	+10.67%	0.455	−11.00%
Attention (gross)	+14.25%	0.629	−9.77%
Price-only (net, 10 bps)	−45.81%	−2.414	−46.34%
Attention (net, 10 bps)	−39.00%	−2.130	−39.53%

Table 3.2: Later evaluation period, next-open entry

Compared with the original close-to-close backtest, entering trades at the following day's open reduced gross cumulative returns from 29.40% to 10.67% for the price-only strategy and from 24.03% to 14.25% for the attention-informed strategy. Gross Sharpe ratios also declined, although attention now outperformed price-only, with 0.629 against 0.455, reversing their original ranking. This is

consistent with slower adjustment in lower-attention followers, but as a single test with both strategies still unprofitable after costs, it does not overturn the main result.

4. Methodology Defense

4.1 Stock Universe and Data Selection

Restricting relationships to companies within the same industry follows Hou (2007), who argues that one firm's returns help predict another's mainly when both are in the same industry, because news affecting an industry takes time to be reflected in all of its firms' prices. This provided an economic basis for examining relationships between companies exposed to similar developments, rather than relying on correlations between unrelated firms. Selecting 10 companies per industry ensured that each group contributed the same number of potential relationships, while keeping the network computationally manageable.

Bloomberg news-publication counts were used as a proxy for investor attention, as Barber and Odean (2008) identify news coverage as a factor that attracts investor attention and influences trading behaviour. This provided a research basis for using publication counts, although they cannot establish whether investors actually read or respond to the news (Da, Engelberg and Gao, 2011). A 20-trading-day average (approximately 1 trading month) was chosen to reduce the influence of individual news spikes while remaining responsive to changing coverage. Alternative 10- and 40-day windows were also tested, detailed in Section 5.3.

4.2 Network construction and trading signals

A 252-trading-day estimation window was selected to provide approximately one year of historical observations, following the use of a 12-month formation period in related stock-pair research by Gatev, Goetzmann and Rouwenhorst (2006). Networks were recalculated every 20 trading days, matching the one-month horizon of the attention measure, so that relationships could be revised as new observations became available without recalculating them daily.

A 20-day return z-score was also used to identify price movements that were unusually large relative to each stock's recent returns. A threshold of two standard deviations was used, consistent with the approach of Guo, Dong and Patterson (2024) to identifying extreme losses. In the development period, 7.4% of stock-days met this threshold, and thresholds of 1.5 and 2.5 were also tested during development to examine whether the results depended on this definition. Signals were weighted by correlation strength to give stronger relationships more capital, with gross exposure capped at 100% to prevent leverage from driving the results.

The strongest 20% of relationships within each industry were retained to exclude weaker historical associations while maintaining a sufficient number of relationships for generating signals. The effect of this choice was examined by testing alternative selections of 10% and 30%. Since relationships were ranked by correlation strength rather than statistical significance, some selected relationships may still be unreliable.

4.3 Development and Evaluation

Following the estimation window, defined in Section 4.2, the first 252 observations provided the history required to construct the initial network. The remaining 2021-2023 observations were used to examine the trading rules and parameter choices, while January 2024-September 2026 was used for a later evaluation with all strategy parameters fixed in advance. A chronological split was used because return predictability found in one sample often weakens in a different sample (McLean and Pontiff, 2016). Moreover, splitting at January 2024 gave similar lengths in development and out-of-sample periods.

4.4 Returns, Execution and Transaction Costs

Closing-price returns were used to generate signals because the hypothesis concerns how prices adjust to one another; they differ from total returns only through small ex-dividend drops (an average absolute daily gap of about 0.01%). Gross total-return indices were used to calculate performance so that dividends received are included. The main backtest assumes that positions based on the closing signal can earn the following close-to-close return, although the signal is only observable at that same close. The next-open test in Section 3.2 avoids this assumption, although it uses open-to-close price returns rather than dividend-inclusive close-to-close returns, so the two backtests differ in their holding periods as well as their entry assumptions.

Transaction costs were modelled at 10 basis points (0.10%) per unit of traded value, informed by Frazzini, Israel and Moskowitz (2012), who report average market-impact costs of 11.21 basis points for large-cap stock trades and a median of 7.98 basis points across all trades. The model deducts costs according to changes in portfolio positions, while the next-open test accounts for both entry and exit.

5. Pitfall Checklist

5.1 Look-ahead bias

Look-ahead bias was addressed by ensuring that trading signals relied on historical information. Networks were constructed using preceding returns, while the 20-day news-attention measure and the mean and standard deviation used for return z-scores were shifted by one trading day. The notebook also asserts that neither strategy holds positions during the 252-day warm-up and confirms that a signal on day t earns the follower's total return on day t + 1. As mentioned in Section 4.4, the main backtest assumes that trades are filled at the same closing price used to compute the signal. The next-open test in Section 3.2 removes this assumption, and the fall in gross returns under that test shows the results were sensitive to execution timing.

5.2 Survivorship bias

The study used a fixed universe of 50 established U.S. equities rather than reconstructing the companies available at each historical date. Since the companies were chosen from firms still listed today, any firm that went bankrupt, was acquired or was delisted between 2020 and 2026 was not included in this study. This creates potential survivorship bias, since excluding unsuccessful companies could overstate historical trading performance. A more rigorous evaluation would use historical company membership records and include delisted securities to assess whether the findings remain consistent.

5.3 Overfitting/data-snooping

Overfitting was examined through nine development-period combinations of shock thresholds (1.5, 2.0 and 2.5) and relationship-selection percentages (10%, 20% and 30%). Attention achieved a higher Sharpe ratio in only four of the nine combinations, while both strategies had negative average Sharpe ratios. Of the 10-, 20- and 40-day attention windows, the 40-day window performed best, achieving a Sharpe ratio of −0.105, but all three remained negative. Additionally, the attention strategy ranked at the 40th percentile among 100 randomly selected, industry-matched networks by Sharpe ratio. In total, nine parameter combinations and two alternative attention windows were tested, none of which replaced the original specification, and attention did not provide a robust improvement.

5.4 Transaction costs and slippage

As mentioned in Section 4.4, transaction costs were evaluated using the 10-basis-point assumption, alongside sensitivity tests at 5 and 20 basis points. Both strategies became unprofitable after costs, including under the lower 5-basis-point assumption, where later-period cumulative returns were −6.97% for price-only and −7.27% for attention. Break-even costs were about 3.9 bps per side for price-only and 3.7 for attention, so the strategies would only be profitable for traders with costs below

roughly 4 bps. This reflects a very high turnover, with annualised turnover in the later period being 241.7× for price-only and 213.0× for attention. Slippage between the signal close and the next available price was examined through the next-open test, detailed in Section 3.2. However, these calculations use fixed proportional costs rather than observed bid-ask spreads or market impact, so actual execution costs could differ.

5.5 Out-of-sample validation

January 2024–September 2026 was held out from all parameter choices: every later-period run used the specification fixed in development, and the parameter, attention-window and random-network tests used development data only. Its only earlier use was an exploratory comparison of lagged correlations, but did not inform any parameter choice. Performance was stronger in this period than in development (see Table 3.1), the opposite of the pattern overfitting would produce.

6. Efficiency Touchpoint

The most computationally expensive step was generating trading signals, which required checking selected relationships for every trading day. Its time complexity was $O(D \times E)$, where D represents trading days and E the number of selected relationships. This became particularly expensive when repeated across 100 random-network simulations. Since E is a fixed 20% of the $k(k-1)$ directed pairs in each industry of k stocks, it grows quadratically with industry size, so doubling each industry to 20 stocks would raise E from 90 to 380.

Caching did not change this step's complexity, but it removed repeated work elsewhere: the 38 historical networks were calculated once rather than rebuilt for every simulation. This reduced the total runtime from 181.36 to 122.65 seconds, a 1.5× speed-up, with identical strategy returns. Network construction itself took only 0.62 seconds, while the remaining signal calculations averaged 1.23 seconds per simulation, which is acceptable for 50 stocks. While processing signals through matrix operations rather than nested loops would improve the scalability, the complexity remains unchanged.

7. Risk Touchpoint

The exposure cap ruled out leverage but not concentration. During the later evaluation period, the strategies held an average of approximately 6.4-6.6 stocks on active trading days, with the largest single-stock positions reaching 57.4% for price-only and 78.4% for attention-informed. This indicates that the portfolios were considerably less diversified than the 50-stock universe might suggest.

The portfolios were also not market-neutral: average absolute net exposure was 31% for price-only and 27% for attention, reaching 100% on some days. A market shock in which lead-lag relationships break down would therefore hit several concentrated, partly directional positions at once; the price-only strategy's two worst daily losses of −3.18% and −2.85%, followed signals formed on 9 and 10 April 2025, during the tariff-driven volatility spike.

8. Reflection

I initially expected that differences in investor attention would help identify which stocks respond to information more quickly, improving the predictive value of historical lead-lag relationships. However, the results challenged this assumption. The attention-informed strategy did not consistently outperform the price-only network, and the performance of both strategies changed considerably between the development and later evaluation periods. Supplementary tests (Section 9 in the notebook) also showed that most later-period gains came from reversal relationships and industry-wide moves rather than the same-direction responses I had expected, so the strategy profited from something other than my proposed mechanism.

The most important lesson was the difference between statistical predictability and practical profitability. Although both strategies generated positive gross returns in the later period, frequent

trading caused these gains to disappear after transaction costs. The next-open test further demonstrated how execution assumptions could change the apparent performance of a strategy. I had initially underestimated how significantly these implementation details could affect the outcome.

The next step would be to test the strategy on a larger, historically reconstructed stock universe to assess whether the findings generalise. I would also investigate more direct measures of investor attention and alternative signal horizons to determine whether predictive relationships persist over longer periods. Before considering real-world implementation, the strategy would require more realistic execution modelling, including bid-ask spreads, market impact and methods to reduce turnover. Only if it demonstrated consistent performance after these costs would scaling to a larger portfolio be justified.

References

Barber, B.M. and Odean, T. (2008) 'All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors', *The Review of Financial Studies*, 21(2), pp. 785–818. Available at: https://doi.org/10.1093/rfs/hhm079.

Brennan, M.J., Jegadeesh, N. and Swaminathan, B. (1993) 'Investment Analysis and the Adjustment of Stock Prices to Common Information', *The Review of Financial Studies*, 6(4), pp. 799–824. Available at: https://doi.org/10.1093/rfs/6.4.799.

Da, Z., Engelberg, J. and Gao, P. (2011) 'In Search of Attention', *The Journal of Finance*, 66(5), pp. 1461–1499. Available at: https://doi.org/10.1111/j.1540-6261.2011.01679.x.

Frazzini, A., Israel, R. and Moskowitz, T.J. (2012) *Trading Costs of Asset Pricing Anomalies*. Working paper. Greenwich, CT: AQR Capital Management. Available at: https://conference.nber.org/conf_papers/f68262/f68262.pdf (Accessed: 20 October 2026).

Gatev, E., Goetzmann, W.N. and Rouwenhorst, K.G. (2006) 'Pairs Trading: Performance of a Relative-Value Arbitrage Rule', *The Review of Financial Studies*, 19(3), pp. 797–827. Available at: https://doi.org/10.1093/rfs/hhj020.

Guo, X., Dong, H. and Patterson, G.A. (2024) 'Equity Returns Around Extreme Loss: A Stochastic Event Approach', *American Business Review*, 27(1), pp. 207–220. Available at: https://doi.org/10.37625/abr.27.1.207-220.

Hou, K. (2007) 'Industry Information Diffusion and the Lead-Lag Effect in Stock Returns', *The Review of Financial Studies*, 20(4), pp. 1113–1138. Available at: https://doi.org/10.1093/revfin/hhm003.

McLean, R.D. and Pontiff, J. (2016) 'Does Academic Research Destroy Stock Return Predictability?', *The Journal of Finance*, 71(1), pp. 5–32. Available at: https://doi.org/10.1111/jofi.12365.

Novy-Marx, R. and Velikov, M. (2016) 'A Taxonomy of Anomalies and Their Trading Costs', *The Review of Financial Studies*, 29(1), pp. 104–147. Available at: https://doi.org/10.1093/rfs/hhv063.